

As a result of the efforts of the OHA, the first Android phone was released in 2008. Called the HTC Dream and popularly known as the G1, the device marked HTC's foray into the Android segment. Several other manufacturers would follow suit, resulting in the market getting flooded with a wide variety of mobile devices, all running Google's Android platform.

What is Android?

Put simply, Android is the sum total of all the software that allows a device to function as a phone.

At the lowest level of abstraction reside the various hardware components like the internal circuitry, the sensors and various radios like the GSM telephony that handles the device's phone operation, and the Wi-Fi,



GPS and Bluetooth radios. On top of this layer is the operating system's kernel. Android uses the free and open source Linux kernel, (originally modelled after the mainline kernel, version 2.6) with some tweaks with regards to power management. These tweaks were submitted back to the mainline kernel, but were rejected due to Google's lack of interest in maintaining them. Moreover, the kernel has been modified to make it compatible with the ARM architecture, known

for its energy efficiency. The next layer consists of three parallel (from an abstraction point of view) components –

1. The libraries written in C that perform various data processing functions, such as SQL lite for managing structured databases, The Open GL ES library for 3D graphics and the Webkit engine for rendering web pages.
2. The core libraries written for Android as well as the Java libraries that deal with the various Java APIs used while writing Android applications
3. Dalvik Virtual Machine, that manages the memory used by various applications and assigns a separate process to each application, so that should an application crash, it won't bring the whole system down with it. Above the DVM resides the Application Framework Layer

The Application framework layer is the highest level of abstraction, containing many higher level libraries such as those used in telephony, data resources and the user interface. All Android applications are built on the Application framework layer. The Android system allows the reuse of code components, thus easing the developers quite a bit of hassle in sharing resources such as data and services between applications.